

MicroPulse

MD Comparison — HSR Exposure

Coach Guide — periodization and match readiness

1. Overview — what is MD Comparison?

MD Comparison — HSR Exposure is a periodization tool that tells you whether each player has accumulated enough high-speed running this week to be ready for the next match. This is not a load overview — it is a READINESS overview.

Think of it as **a mirror that compares what a player has done this week to what the match will demand of him**. Too little HSR before a match = the player is not accustomed to the speeds he'll be forced to run = high injury risk. Too much HSR = the player is pre-fatigued when he steps onto the pitch.

The 95% rule (Buchheit 2018)

Players who reach $\geq 95\%$ of their typical MD HSR by MD-1 (the day before the match) enter the match prepared. Players who reach less than 80% across the whole cycle are at significantly elevated acute-injury risk on match day.

Where do you find this?

- Page: /coach/load-intelligence — the bottom card on the page.
- Top-tab: "MD Comparison" tab in the coach dashboard (where applicable).
- Visible to all Catapult plans — both Lite and Pro / Premium.

2. The science

2.1 — Buchheit 2018 (95% rule)

Finding: "Players need to be exposed to the speeds they'll be required to run at on match day, repeatedly, in training." Players who reach 95-110% of their MD HSR by MD-1 are as ready as the system can verify. Players who fall under 80% have a significant increase in acute-injury risk on match day (especially hamstring strains in the final 15 minutes).

2.2 — Stares 2018 (chronic high-speed exposure protects)

Finding: A longer history of regular high-speed exposure in training is protective in elite football. This is not just about this week — it's about the habit. The right response when a player is under-exposed is not to drop high-speed work. It is to add it back gradually so chronic load returns to its protective level.

Why this tool is separate from HSR Intelligence

HSR Intelligence (Malone 2017) is about chronic spike risk — is this player ramping HSR up too fast? MD Comparison is about match readiness — has this player accumulated the right exposure for the next match? These are different questions. The same player could be GREEN on HSR ACWR (clean chronic ramp) but RED UNDER on MD Comparison (this week has been too light for

the match on Sunday).

3. Column-by-column glossary

The card shows a table with one player per row, sorted by risk severity (UNDER first — those are the players who need the most attention).

Column	What it measures	How to read it
Player	Player name and position.	Self-explanatory — no calculation.
HSR 7d (m)	Total high-speed running over the past 7 days (metres).	Compared against the MD baseline in the next column.
MD baseline (m)	The player's personal match-day demand. Two methods: (1) median HSR from matches the player started ≥30 min in the last 90 days, or (2) synthetic — top 10% of his own sessions if no matches have been played yet.	This is a personal benchmark — not a team average. 1500 m for a midfielder vs 800 m for a goalkeeper is all normal.
% of MD	$\text{HSR 7d} / \text{MD baseline} \times 100$. This is the headline number.	<80% = UNDER (red), 80-95% = WATCH (yellow), 95-110% = READY (green), >110% = OVER (yellow).
Samples	How many matches / sessions were used to compute the baseline. "3 match" = baseline from 3 matches. "15 sess" = synthetic baseline from 15 sessions.	Fewer samples = less reliable baseline. Trust a "5 match" baseline more than a "1 match" one.
Status	Combined flag for the player. Colour-coded red / yellow / green.	What you read first — it tells you priority.

Strategy badge — at the top of the card

At the top of the card you'll see which strategy was used for the team:

Match-derived baseline (green badge): At least one player has played a match in the last 90 days. The baseline uses the median HSR distance from matches each player started in. This is the gold standard — the canonical Buchheit / Stares method.

Synthetic baseline (blue badge): No matches played in the 90-day window. Uses the top 10% of each player's own sessions as a proxy for MD demand. Pre-season state. Less accurate, but it keeps the tool useful from day 1 of pre-season instead of going dark for 6 weeks.

4. Status flags — what each one means

Status	% of MD	What it means	What you should do
UNDER	<80%	The week has been too light for the upcoming match demand. The muscle is not accustomed to the speeds it will be forced to run.	Add HSR to the MD-1 session (positional drills, possession on a larger pitch, or short 30-50 m sprints). Or: reduce match minutes if you can't make up the volume.
WATCH	80-95%	Almost in the safe zone. Needs a	The MD-1 session should add 200-

Status	% of MD	What it means	What you should do
		top-up.	400 m of HSR to bring the player into the ready band.
READY	95-110%	Optimal. The week has matched the expected match demand.	No specific action. Keep the normal MD-1 plan (taper).
OVER	>110%	Too much HSR this week. Player will be pre-fatigued unless MD-1 is light enough.	Make the last session lighter (recovery / technical). Avoid HIT, possession, or full-speed sprints in MD-1.

5. Workflow — when to use this

MD-3 / MD-2 (third / second day before the match) — week-plan adjustment

- Open MD Comparison on Tuesday (if the match is on Saturday).
- Which players are UNDER? They need an MD-2 or MD-1 session with extra HSR — e.g. an SSG with more defensive sprints, or positional sprints.
- Who is OVER? Plan a personally lighter MD-1 (scheme work, set pieces, recovery).

MD-1 (day before the match) — final readiness check

- Re-check after yesterday's session. Have UNDER players moved into WATCH or READY?
- If still UNDER on MD-1: add 10-15 min of positional sprints at the end of the session. Don't let a player enter a match below 80%.
- Check match samples — if a player has a synthetic baseline with fewer than 5 sess, treat it cautiously. Also ask the player: "do you feel ready?"

After the match — match-baseline calibration

- After every match a new match data point is automatically added to the baseline. Synthetic teams gradually become match-derived after 1-2 matches.
- Look at who was OVER going into the match — were they high-HSR pre-match and possibly over-tired? Watch them on the next MD-1.

6. Frequently asked questions

Why is one player showing "—" in MD baseline?

It means the player has no usable HSR history in the last 90 days — a new player who hasn't yet trained or played. Wait until they've trained for at least 1 week.

Why is my baseline different from the team average?

Because the baseline is PERSONAL. A goalkeeper might have a 600 m baseline; a midfielder 1700 m. Holding the goalkeeper to 1700 m would mark him as ALWAYS UNDER. A personal benchmark is essential for accurate risk assessment.

What if a player is READY here but RED on HSR Intelligence?

This is entirely possible and is a good example of why both tools exist. MD Comparison tells you the exposure this week matches the MD demand. HSR Intelligence tells you the ramp-up over the last 4 weeks has been too steep (ACWR > 1.5). Both can be true at the same time. It usually means: "the player is ready for the match, but he is not accustomed to this much volume" — which is exactly what a spike pattern looks like.

What happens with DNP (did not play) matches?

Excluded from the baseline. So are matches where the player played <30 min — they are not meaningful match exposure. Only matches with ≥30 min count.

Can I tune the 95% threshold?

No — taken directly from Buchheit 2018. Changing it would lose the tool's evidence base. If you want more sensitive warnings, HSR Intelligence (Malone 2017) has lower thresholds (1.3 ACWR = yellow).

When is the strategy "match" vs "synthetic"?

Synthetic is used in pre-season and early in the season when there are no matches in the 90-day window. After the 1st or 2nd match of the season the strategy will switch to match. Mid-season, it should always be match-derived.

7. Limitations

- The synthetic baseline (pre-season) is less accurate than match-derived. Top-decile sessions don't always reflect actual match demand — some hard sessions exceed any match.
- Looks at HSR only. Sprint distance, sprint count, and max-V are not included — you need to cross-check HSR Intelligence for that dimension.
- Uses a 90-day window for the match baseline. If a player has played only 1 match in 90 days, the median is from that single match (low stability).
- Baseline reflects what the player has historically done — not what the match will actually demand. If the player has consistently underperformed his role (e.g. in poor matches), the baseline will be too low.
- Shows HSR exposure only. Tactical changes (e.g. a switch to a higher pressing system) that will demand more HSR are not visible — you need to factor those in manually.

8. References

Buchheit M. (2018). Managing high-speed running load in professional soccer players: the benefit of high-intensity interval training supplementation. *Sport Performance & Science Reports*, 53.

Stares J, Dawson B, Peeling P, Drew M, Heasman J, Rogalski B, Colby M. (2018). Identifying high risk loading conditions for in-season injury in elite Australian football players. *Journal of Science and Medicine in Sport*, 21(1), 46-51.

Malone S, Owen A, Mendes B, Hughes B, Collins K, Gabbett TJ. (2017). The acute:chronic workload ratio in relation to injury risk in professional soccer. *Journal of Science and Medicine in Sport*, 20(6), 561-565.